

Brazilian E-Commerce Business Analysis

VirtuBox Data Analyst Assessment

Olist Brazilian E-Commerce Public Dataset

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2. Data & Methodology

- Dataset: Olist Brazilian E-Commerce Public Dataset.
- The assessment allows large public datasets, multiple related files, and complex business data.
- The Olist data contains multiple related datasets covering orders, items, products, customers, sellers, payments and reviews.
- Tools used: Python, Pandas and Google Colab for data loading, cleaning, validation and exploratory analysis.
- Checks performed included missing values, duplicate records, order status, customer/seller states, product categories, payment methods and review scores.

3. Key Findings — Executive Summary

- 99,441 unique orders were present in the orders dataset.
- 112,650 items were recorded in the order-items dataset.
- Total product-price revenue analysed: 13,591,643.70.
- Average order value based on product price: 136.68.
- 112,650 items across 99,441 orders equals approximately 1.13 items per order, indicating an opportunity to increase basket size.
- Payment and review distributions provide useful signals for checkout optimization and customer-experience improvement.

4. Deep-Dive Insight — Basket Size Opportunity

- The dataset contains 112,650 item records across 99,441 unique orders.
- This corresponds to approximately 1.13 items per order.
- Management opportunity: encourage relevant cross-sells, bundles and product recommendations.
- Why it matters: increasing the number of items per order can improve revenue per customer without requiring the same level of new-customer acquisition.
- Success measures: items per order, average order value, bundle conversion rate and revenue per order.

5. Recommendations

- Priority 1 — Increase basket size: introduce bundles and cross-sell recommendations. Owner: E-commerce/Product team. Measure: items/order and AOV.
- Priority 2 — Improve customer experience: investigate low review scores by product, seller and order characteristics. Owner: Customer Experience/Operations. Measure: average review score and low-score rate.
- Priority 3 — Optimize checkout: monitor the most-used payment methods and make the payment journey reliable and simple. Owner: Product/Payments team. Measure: payment mix, successful checkout rate and order conversion.

6. Expected Business Impact & Measurement

- Higher basket size can increase revenue per existing order.
- Better customer experience can support satisfaction, repeat purchase and seller quality management.
- A smoother payment experience can reduce friction at checkout.
- Management should track AOV, items/order, review-score distribution, low-score rate, payment-method performance and conversion over time.
- Targets should be established after baseline measurement and controlled testing rather than assumed in advance.

7. Limitations & Next Steps

- The dataset is historical and may not represent current customer behaviour or market conditions.
- Missing values and incomplete product/category information can affect some analyses.
- The revenue figure used here is based on item price and is not the same as profit or margin.
- Observational data does not prove causation; relationships should be validated with experiments or deeper analysis.
- Next steps: build the required interactive Looker Studio dashboard, segment findings by category/seller/customer geography, and validate recommendations with A/B testing or operational pilots.